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| United States Patent | 12403550                |
| Kind Code            | B2                      |
| Date of Patent       | September 02, 2025      |
| Inventor(s)          | Anasenzl; Manuel et al. |

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### Method for producing a functional structure and component

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#### Abstract

A method includes producing a functional structure on an aluminum surface with a local laser treatment of an aluminum surface. The local laser treatment is carried out with a pulsed laser system having a pulse duration of from 10 ns to 100 ns. The average power of the pulsed laser system is less than 5 kW.

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**Appl. No.:** 17/396251

**Filed:** August 06, 2021

#### Prior Publication Data

| Document Identifier | Publication Date |
|---------------------|------------------|
| US 20210362275 A1   | Nov. 25, 2021    |

#### Foreign Application Priority Data

|    |                   |               |
|----|-------------------|---------------|
| DE | 10 2019 103 130.3 | Feb. 08, 2019 |
|----|-------------------|---------------|

#### Related U.S. Application Data

continuation parent-doc WO PCT/EP2020/051700 20200124 PENDING child-doc US 17396251

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#### Publication Classification

**Int. Cl.: B23K26/12** (20140101); **B23K26/32** (20140101); **B23K26/356** (20140101);  
**B23K103/10** (20060101); **C21D10/00** (20060101)

**U.S. Cl.:**

**CPC B23K26/356** (20151001); **B23K26/1224** (20151001); **B23K26/125** (20130101);  
**B23K26/32** (20130101); **C21D10/005** (20130101); B23K2103/10 (20180801)

## Field of Classification Search

**CPC:** B23K (26/082); C22F (1/04)

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## References Cited

### U.S. PATENT DOCUMENTS

| Patent No.   | Issued Date | Patentee Name   | U.S. Cl.   | CPC             |
|--------------|-------------|-----------------|------------|-----------------|
| 4972061      | 12/1989     | Duley           | 427/586    | C21D 1/09       |
| 2010/0197116 | 12/2009     | Shah            | 219/121.68 | B23K<br>26/083  |
| 2011/0290915 | 12/2010     | Sugiyama        | 219/121.64 | B23K<br>26/0823 |
| 2014/0349137 | 12/2013     | Brandl et al.   | N/A        | N/A             |
| 2014/0356578 | 12/2013     | Brandl          | 428/141    | B23K<br>26/1224 |
| 2015/0202712 | 12/2014     | Seghi et al.    | N/A        | N/A             |
| 2016/0201184 | 12/2015     | Seghi et al.    | N/A        | N/A             |
| 2016/0265570 | 12/2015     | Sabau et al.    | N/A        | N/A             |
| 2017/0136668 | 12/2016     | Kose et al.     | N/A        | N/A             |
| 2017/0291259 | 12/2016     | Imoto           | N/A        | B23K<br>26/323  |
| 2018/0369956 | 12/2017     | Anasenzl et al. | N/A        | N/A             |

### FOREIGN PATENT DOCUMENTS

| Patent No.      | Application Date | Country | CPC          |
|-----------------|------------------|---------|--------------|
| 106552997       | 12/2016          | CN      | N/A          |
| 106715073       | 12/2016          | CN      | N/A          |
| 108367383       | 12/2017          | CN      | N/A          |
| 19509497        | 12/1995          | DE      | B23K 26/40   |
| 102009054394    | 12/2010          | DE      | B23K 26/0006 |
| 10 2011 121 545 | 12/2012          | DE      | N/A          |
| 10 2011 121 546 | 12/2012          | DE      | N/A          |
| 10 2015 000 179 | 12/2015          | DE      | N/A          |
| 1 854 903       | 12/2006          | EP      | N/A          |
| 2 669 041       | 12/2012          | EP      | N/A          |
| 3034224         | 12/2015          | EP      | B23K 1/0012  |
| 5-115990        | 12/1992          | JP      | N/A          |
| 2016129942      | 12/2015          | JP      | B23K 26/324  |

|                |         |    |     |
|----------------|---------|----|-----|
| WO 2007/142610 | 12/2006 | WO | N/A |
| WO 2015/172797 | 12/2014 | WO | N/A |
| WO 2018/188875 | 12/2017 | WO | N/A |

## OTHER PUBLICATIONS

Translation of DE-19509497-C1 (Year: 1996). cited by examiner

Translation of DE-102009054394-A1 (Year: 2011). cited by examiner

Translation of JP-2016129942-A (Year: 2016). cited by examiner

International Search Report (PCT/ISA/210) issued in PCT Application No. PCT/EP2020/051700 dated Jun. 17, 2020 with English translation (five (5) pages). cited by applicant

German-language Written Opinion (PCT/ISA/237) issued in PCT Application No.

PCT/EP2020/051700 dated Jun. 17, 2020 (five (5) pages). cited by applicant

German-language Search Report issued in German Application No. 10 2019 103 130.3 dated Nov. 14, 2019 with partial English translation (12 pages). cited by applicant

Heckert A., "Pulsed Laser Surface Pre-Treatment of Aluminium to Join Aluminium-Thermoplastic Hybrid Parts," Lasers in Manufacturing Conference (LiM) 2015 (12 pages). cited by applicant

Chinese Office Action issued in Chinese Application No. 202080006880.7 dated Feb. 7, 2022 (11 pages). cited by applicant

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## Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of PCT International Application No. PCT/EP2020/051700, filed Jan. 24, 2020, which claims priority under 35 U.S.C. § 119 from German Patent Application No. 10 2019 103 130.3, filed Feb. 8, 2019, the entire disclosures of which are herein expressly incorporated by reference.

### BACKGROUND AND SUMMARY OF THE INVENTION

(1) The present disclosure relates to a method for producing a functional structure on an aluminum surface, to a component comprising a functional structure, and to a use of a pulsed laser system.

(2) For the material connection of two cast aluminum components, for example, it is necessary to treat the surface of the components beforehand in order to achieve a sufficient strength of the joint connection in the subsequent joining process. To this end, inter alia, chemical processes are known from the prior art, for example chemical washing or pickling. Passivation of the surface and the provision of a cathodic dip coating or anodizing processes are also known in this context. Often, however, the known approaches are elaborate in terms of process technology and environmentally unfriendly, and entail high operating and maintenance costs. DE 10 2011 121 546 A1 therefore proposes a method for producing a metal or metal alloy surface or metal oxide or metal alloy oxide layer on a surface, operation being carried out with a pulsed laser system. It has, however, been found that although oxide layers may be produced or provided on aluminum surfaces with such laser systems, they are often for example insufficiently thick or not homogeneous enough to be suitable for a subsequent joining process. To this extent, the problem of operating or configuring the pulsed laser system in a suitable way exists.

(3) It is therefore an object of the present disclosure to provide a method for producing a functional structure on an aluminum surface, a component comprising a functional structure, and a use of a pulsed laser system, which overcome the aforementioned disadvantages.

(4) This object is achieved by a method according to an embodiment of the invention, a method for producing a functional structure on an aluminum surface comprises the step: at least local laser treatment of an aluminum surface in order to produce a functional structure on the aluminum surface or the aluminum alloy surface, wherein the laser treatment is carried out by means of a pulsed laser system having a pulse duration of 10-100 ns and wherein an average power of the laser system is less than 5 kW. Other features and advantages will be explained below with reference to the following figures, in which:

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## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is an aluminum surface during laser treatment to form a functional structure;
- (2) FIG. 2 is the component of FIG. 1 and a joining partner; and
- (3) FIG. 3 is the component and the joining partner, cf. to FIG. 2, after joining via the functional structure.

### DETAILED DESCRIPTION OF THE DRAWINGS

- (4) Surprisingly, it has been found that the functional structure **12** produced in this way entails the formation of an optimal oxide layer. In this case, the oxide layer is in particular very homogeneous and much thicker than has been known. Thus, the thicknesses of oxide layers on AlMgSi alloys (without any laser treatment) is less than 3 nm. With the aforementioned laser treatment, thicknesses of more than 5 nm can be achieved. According to one preferred embodiment, the aforementioned surface is the surface of a component **1** made for example of an aluminum-magnesium (Al—Mg) alloy preferably an aluminum-magnesium-silicon alloy or an aluminum-magnesium-silicon-magnesium (Al—Mg—Si—Mg) alloy. One preferred material is, in particular, AlSi10MnMg. The component **1** is preferably a cast component **1**, according to one embodiment for example a diecast component **1**.
- (5) According to one embodiment, the method comprises the step: material connection of the aluminum surface **10** to a joining partner **2** by means of the functional structure **12**.
- (6) The joining partner **2** is not restricted in detail and may consist of any desired materials, in particular metal materials, for example steel or aluminum or a metal alloy. As an alternative, plastics or fiber composites may be used as a material for the joining partner **2**.
- (7) According to one embodiment, the method comprises the step: material connection by adhesive bonding, welding or soldering.
- (8) The method has been found to be particularly advantageous in combination with an adhesive bonding process. The adhesives conventionally used adhere particularly well on the laser-treated modified surface topology of the region to be joined, i.e. in particular the functional structure **12**, of the aluminum surface **10**.
- (9) Expediently, the functional structure **12** forms or comprises an oxide layer with a thickness in a range of about 5 nm to 700 nm, preferably from 5 nm to 200 nm. In other words, the method may also be referred to as a method for producing an oxide layer on an aluminum surface **10**. By the laser treatment and the input of heat resulting therefrom, the surface is cleaned and modified in such a way that a very uniform oxide layer is formed thereon.
- (10) Particularly preferably, the laser system **20** has an average power of between 1 and 3 kW, or between 1 and 2 kW, or the laser treatment is carried out with this average power. The following method step has been found to be particularly advantageous in this case: focusing the laser beam onto a diameter of about 500 to 3000  $\mu\text{m}$ . Further preferred diameters lie in a range of about 900 to 1800  $\mu\text{m}$ .
- (11) This, particularly with the aforementioned average powers, provides an optimal energy input which leads to the formation of the functional structure **12**, or the oxide layer.

(12) According to one embodiment, the method comprises the step: laser treatment with a pulse duration of 20-40 ns.

(13) This range has been found to be particularly advantageous in terms of the production of the functional structure **12**. In this context, it should also be mentioned that a square beam shape is preferably used as the beam shape, although any other desired beam shapes may also be used. The square beam shape has, in particular, the advantage that in particular uniform processing without overlap regions is made possible when scanning with the laser. Particularly preferably, the laser treatment is carried out at the focal position and with a large Rayleigh length, preferred values for the Rayleigh length lying between 10 and 25 mm. The surface treatment may thus be made more robust, and is for example more tolerant in relation to focal position variations.

(14) For the surface treatment, a wavelength of the laser system **20** of 1000-1100 nm has been found to be particularly advantageous since the pulse waves can thus be absorbed and converted very well by the aluminum material. The wavelength is particularly advantageously 1030 and 1064 nm.

(15) Furthermore, the pulse frequency is advantageously 10-40 kHz, and in particular 10-30 kHz. In this way, the surface topology and surface tension of the cast aluminum component **1** to be joined are modified optimally with a reduced energy outlay, so that the adhesion to the joining partner **2** can be increased.

(16) Advantageously, no chemical surface treatment steps are used. Correspondingly, the method advantageously comprises the step: connection immediately after the laser treatment, or joining immediately after the laser treatment.

(17) Advantageously, no further method or process steps are thus required after the laser treatment. This also applies in respect of any pretreatment before the laser treatment, which is likewise not necessary. In a particular case, however, it is readily possible to clean the surface to be treated beforehand, optionally even mechanically.

(18) According to one embodiment, the method comprises the step: laser treatment in an atmosphere, the atmosphere being a vacuum, an ambient atmosphere or an inert gas or gas mixture.

(19) According to one embodiment, the aluminum surface **10** belongs to an aluminum component **1**, in particular a cast aluminum component **1**. The aluminum material used according to the method is not restricted in detail, so long as aluminum constitutes the main component in percent by mass, expressed in terms of the total weight of the component **1**. Particularly advantageously, an aluminum content in the cast aluminum component **1** to be connected to the joining partner **2** is at least 70 percent by mass, expressed in terms of the total weight of the cast aluminum component **1**.

(20) The disclosure also relates to a component **1**, in particular an aluminum component **1**, in particular a cast aluminum component **1**, comprising a functional structure **12** produced by the method according to the disclosure, wherein the functional structure **12** comprises an oxide layer having a thickness of about 5-700 nm. This entails optimal adhesive bonding and corrosion resistance. The functional structure **12**, or laser-treated surface, provides any joining partners **2** with surface-active groups for bonding and is distinguished by an increased oxygen concentration. In particular, the oxide layer is extremely homogeneous and uniform. The uniformity of the oxide layer is distinguished by a "mirrored" component **1** surface in the laser-treated regions.

Advantageously, the zone of fusion or zone of thermal influence into the surface, or into the component **1**, is very small, for example up to at most 10  $\mu\text{m}$ .

(21) The disclosure furthermore relates to the use of a pulsed laser system **20** having a pulse duration of 10-100 ns and an average power of between 1 and 5 kW for producing a functional structure **12**, or an oxide layer, on an aluminum surface **10**, in particular of a cast aluminum component **1**. The advantages and features explained in connection with the method or with the component **1** apply similarly and correspondingly for the use.

(22) The laser system **20** used may for example comprise a beam source, a cooling device for the beam source, a guide fiber from the beam source to a scanner, and a scanner (for the beam

deflection), in which case the scanner may be arranged on a mobile device, for example on a robot. (23) The foregoing disclosure has been set forth merely to illustrate the invention and is not intended to be limiting. Since modifications of the disclosed embodiments incorporating the spirit and substance of the invention may occur to persons skilled in the art, the invention should be construed to include everything within the scope of the appended claims and equivalents thereof.

## Claims

1. A method comprising: producing a functional structure on an aluminum surface with a local laser treatment of an aluminum surface, wherein the local laser treatment is carried out with a pulsed laser system having a pulse duration of from 20 ns to 40 ns, an average power of the pulsed laser system is less than 5 kW, the functional structure comprises an oxide layer formed on the aluminum surface, the oxide layer having a thickness of from 60 nm to 700 nm, a wavelength of the laser system is between 1000 nm and 1100 nm, a zone of fusion or zone of thermal influence into the aluminum surface is at most 10  $\mu$ m, and the pulsed laser is applied at a focal position and with a Rayleigh length lying between 10 and 25 mm.
  2. The method according to claim 1, further comprising: forming a material connection between the aluminum surface and a joining partner, wherein the material connection is formed via the functional structure of the aluminum surface.
  3. The method according to claim 2, wherein the material connection is formed by adhesive bonding, welding, or soldering.
  4. The method according to claim 1, wherein the pulsed laser system has an average power of between 1 kW and 2 kW.
  5. The method according to claim 4, further comprising: focusing the laser beam onto a diameter of from 500  $\mu$ m to 3000  $\mu$ m.
  6. The method according to claim 1, further comprising: establishing the material connection after the local laser treatment.
  7. The method according to claim 6, wherein the local laser treatment is applied in an atmosphere, the atmosphere being a vacuum, an ambient atmosphere, or an inert gas or gas mixture.
  8. The method according to claim 7, wherein the aluminum surface belongs to a cast aluminum component.
  9. The method according to claim 1, wherein a beam of the pulsed laser system has a square shape.
  10. The method according to claim 1, further comprising: focusing the laser beam onto a diameter of from 900  $\mu$ m to 1800  $\mu$ m.
  11. The method according to claim 1, wherein the wavelength of the laser system is between 1030 nm and 1064 nm.
  12. A method for producing a functional structure comprising: applying a laser with a pulsed laser system having a pulse duration of from 20 ns to 40 ns and an average power of between 1 kW and 5 KW, to thereby produce the functional structure on an aluminum surface, wherein the functional structure comprises an oxide layer formed on the aluminum surface, the oxide layer having a thickness of from 60 to 700 nm, and a wavelength of the laser system is between 1000 nm and 1100 nm, a zone of fusion or zone of thermal influence into the aluminum surface is at most 10  $\mu$ m, and the pulsed laser is applied at a focal position and with a Rayleigh length lying between 10 and 25 mm.
  13. The method according to claim 12, further comprising: focusing the laser beam onto a diameter of from 900  $\mu$ m to 1800  $\mu$ m.
  14. The method according to claim 12, wherein the wavelength of the laser system is between 1030 nm and 1064 nm.
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